AI Implementation in NHS: Overcoming Challenges for Smart Healthcare Systems

Business Consulting (MANP005)

By

Muhammad Imran (3457775)

Date: 15 April 2025

Executive Summary

In this report my objective is to work as a business analyst to highlight how AI can be successfully implemented in the National Healthcare System (NHS) in the UK to enhance its productivity and reduce waiting time for patients. We will talk about the history of NHS, its structure and services from the past till today and analyze what causes it to fail the Uks best healthcare system. As of the past, there were lots of financial losses when attempts were made to digitalize the healthcare system in the UK. This happened because of poor governance and inappropriate budget allocation. In this report we will find the ways that led to failure in implementing AI in the NHS and how these issues can be addressed. Also, we would go through multiple issues in issue tree, analyze these in detail, address each issue and provide proper recommendations as an analyst on how we can be successful in our objective.

By adopting frameworks like the McKinsey 7-Step Framework and the Quadruple Helix Model, we will identify the key issues and challenges faced in the NHS, analyze data to inform decision-making, and provide a comprehensive plan for successful AI implementation. Additionally, we explore how investing in digital infrastructure could help alleviate the burden on the workforce, reduce financial losses, and improve patient care. We will also dive deep into the detailed study of how we can enhance the digital infrastructure of NHS and let AI do most of the administrative tasks to reduce the burden on the medical workforce. The same way we will analyze if we just spend 10-15% of the total NHS budget on developing the infrastructure we can reduce waiting time and financial losses that can be re-invested to boost productivity.

Contents

AI Implementation in NHS: Overcoming Challenges for Smart Healthcare Systems 1

1 Introduction and Background 4

2 NHS Infrastructure across the UK 5

2.1 NHS England 5

2.2 NHS Northern Ireland 5

2.3 NHS Scotland 5

2.4 NHS Wales 6

3 Project Approach and Methodology 7

3.1 Applications for McKinsey 7-Step Framework 7

3.2 Integration of Quadruple Helix Model 8

4 Issue Tree and Problematization 9

4.1 HR Challenges 9

4.2 Financial Constraints 10

4.3 Structural Issues 10

5 Findings and Discussions 10

5.1 Why AI Adoption is Lagging 10

5.2 Strategic Areas for Successful AI Implementation 11

6 Actionable Recommendations and Conclusions 12

6.1 Strengthening Technology Infrastructure 12

6.2 Workforce Readiness and Training 12

6.3 Addressing Regulatory and Ethical Barriers 12

6.4 Ensuring Financial Sustainability and AI Investment 13

6.5 Enhancing Public Trust and Patient Engagement 13

7 References 14

1 Introduction and Background

The National Health Service (NHS) was established in July 1948 by the UK government under Prime Minister Clement Attlee, along with Aneurin Bevan, the then Minister of Health, who led the initiative. The NHS was a publicly funded healthcare system ranging from hospitals to GPs and pharmacists, aiming to provide free medical services to the residents of the United Kingdom. Hence, at that time, the UK became the first Western country to deliver excellent healthcare free of charge. The core principles of this service were equality and care accessible to everyone of all ages. (Gov.uk, 2023).

Before the NHS, people used to get treatments from a mix of local welfare communities, the unregulated private sector, and charities. But this system was not advanced in its infrastructure, and there were gaps between the workforce and the middle class struggling with rising medical costs. The NHS faced little resistance in its formation because it aimed to offer healthcare services through an organisation that prioritized health over financial considerations. (Huddersfield University, 2018). It was then divided into three main components to provide targeted services effectively.

The first and most important sector is the General Medical Service, known for providing primary health care by GPs. This medical service has been the largest part of the NHS since 1948 and has a large workforce. Over 90% of all types of illness were managed by GPs. The second major service is the National Hospital Service good which provides secondary health care by specialists and hospital nurses. Whereas the third and last Public Health Service is meant for promoting good health and preventing diseases usually provided by care physicians and visiting faculty.

2 NHS Infrastructure across the UK

Now let's analyze in detail the infrastructure of the NHS and how it works. The NHS is primarily subdivided in its governance and organizational structure into four parts: England, Northern Ireland, Scotland, and Wales. (Barrett, Freeman, and King, 2022). Each country has its own NHS structure due to devolved health powers, but they all focus on primary care systems to improve health quality across the UK. These four are discussed below:

2.1 NHS England

NHS England operates through Integrated Care Systems (ICSs), which improve healthcare services by coordinating NHS organizations, local authorities, and stakeholders. Integrated Care Boards (ICBs) replaced Clinical Commissioning Groups in 2022, overseeing service commissioning and performance. They work with hospitals, GPs, and community providers to enhance patient care.

Integrated Care Partnerships (ICPs) and Health & Wellbeing Boards (HWBs) tackle local health issues, while Place Boards (PBs) focus on targeted healthcare initiatives. Primary Care Networks (PCNs) serve 30,000–50,000 people, improving accessibility, diagnostics, and remote care. This structured approach ensures a more efficient NHS.

2.2 NHS Northern Ireland

In NHS Ireland, the Department of Health (DoH) oversees social and health care, which replaced HSCB in 2022 to improve service delivery. Local Commissioning Groups (LCGs) assess regional healthcare needs, while Integrated Care Partnerships (ICPs) enhance services for chronic conditions and elderly care.

The Public Health Authority (PHA) promotes health and tackles inequalities to ensure care quality. The Patient and Client Council (PCC) advocates for patients. General Practice Services (GPS) manage primary care, referring cases to secondary care when needed. Health and Social Care Trusts (HSCTs) oversee healthcare delivery across Northern Ireland, ensuring efficient and high-quality services.

2.3 NHS Scotland

In NHS Scotland, we have Regional Boards that manage healthcare delivery through 14 health boards, overseeing hospitals, community services, and district nursing. Integrated Joint Boards (IJBs), comprising 31 boards, handle strategic planning for integrated health care with multiple council inputs, NHS directors and representatives.

The Health and Social Care Partnerships (HSCPs) ensure service delivery across 31 regions, covering adult social care, primary healthcare, and emergency hospital care. GPs and General Practice Clusters, organized into groups of 5–8 practices, focus on quality improvement and local healthcare planning, replacing the Quality and Outcomes Framework (QOF).

2.4 NHS Wales

When we talk about the NHS services in Wales, it's a great part of the discussion. In NHS Scotland, we have National Boards oversee key healthcare services, including Ambulance Services for emergency care, Velindre NHS Trust for cancer treatment and blood services, and Public Health Wales, which focuses on health protection and reducing inequalities.

GPs and General Practice Clusters operate across 64 primary care clusters, serving 30,000–50,000 people. These clusters deliver localized healthcare in collaboration with pharmacies and other primary care providers.

3 Project Approach and Methodology

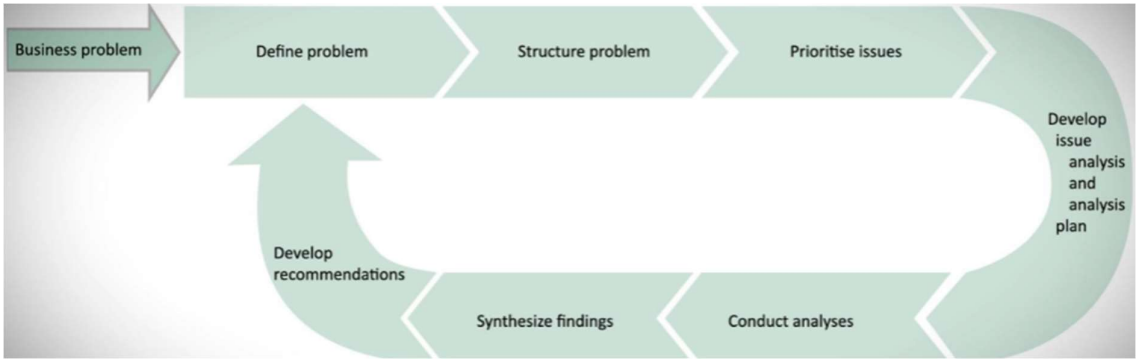
To ensure the credibility and reliability of findings, this research relies upon a mix of primary and secondary sources of information. Generally, the primary source of data was gathered through surveys and structured interviews with NHS healthcare professionals, including doctors, nurses, IT specialists, and administrators. These interactions will help assess the readiness for AI adoption within NHS operations. Additionally, questionnaires will be distributed to patients to gauge public perception of AI-driven healthcare solutions, ensuring that both professional and patient perspectives are incorporated into the study.

Secondary source of data includes NHS Digital reports on financial spending, patient flow efficiency, and AI implementation trials. These reports provide valuable insights into the existing use of technology within NHS systems and the financial viability of further AI integration. To process and interpret the collected data, I employed a combination of data analysis techniques. Descriptive statistics can be used to examine NHS performance, patient waiting times, and the impact of AI on reducing administrative tasks. (Thornton et al., 2024). A comparative analysis must conduct to benchmark NHS AI adoption against other global healthcare systems, identifying best practices and areas for improvement. A cost-benefit analysis should be undertaken to evaluate the financial returns on AI investments, ensuring that proposed solutions are economically viable.

I conducted research over 12 weeks, with each phase carefully structured to ensure thorough analysis and comprehensive findings. The first two weeks was more focused on a literature review, examining existing research on AI in healthcare and NHS performance trends. Weeks three and four involved data collection from NHS Digital reports and AI case studies, laying the groundwork for empirical analysis. In weeks five and six, surveys and recoded interviews with NHS professionals was analysed to gather firsthand insights into AI adoption challenges and opportunities. Weeks seven and eight were about dedicated to quantitative analysis of AI efficiency within NHS workflows, followed by a comparative analysis of AI healthcare models in other countries during weeks nine and ten. In eleventh week, I was used to draft findings and discuss AI adoption challenges, ensuring all insights are coherently presented. Finally, in week twelve, the research got completed in developing recommendations and finalizing the report, providing actionable insights into successful implementation of AI in healthcare system.

3.1 Applications for McKinsey 7-Step Framework

This framework provides a structured, step-by-step approach to solving business problems in consulting. It begins by clearly defining and breaking down the problem into manageable parts, prioritizing key issues, and developing an analysis plan. After that, to conduct data-driven analyses, synthesize findings, and formulate actionable recommendations. This method ensures a logical, evidence-based approach to decision-making, making it valuable for addressing strategic, operational, and financial challenges in businesses.



Source: - Lecture 1: Introduction to Business Consulting

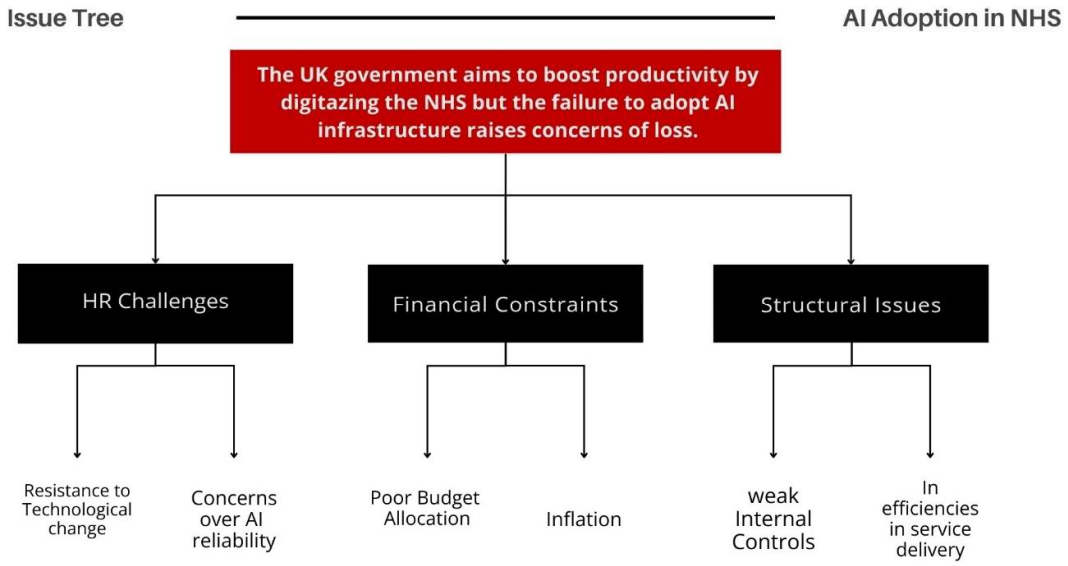
3.2 Integration of Quadruple Helix Model

The Quadruple Helix Model emphasizes that innovation is more than just an interaction between government, industry, and academia. In this innovative framework, there are four key stakeholders: government, industry, academia, and civil society. It highlights the importance of collaboration between these groups to foster innovation, economic growth, and societal progress. The government sets policies and regulations, industry brings innovations to market, academia conducts research and development, and civil society provides valuable feedback, needs, and values, ensuring that innovations are inclusive, sustainable, and address real-world challenges. (Young, D., 2023)

In this assignment, the Quadruple Helix Model can be implemented by fostering collaboration between the NHS (as the government/healthcare provider), healthcare technology companies (industry), research institutions (academia), and patients or the public (civil society). By engaging these four groups, it can be ensured that innovations in healthcare, whether in AI, digital tools, or patient care, are not only technologically advanced but also aligned with patient needs and regulatory health standards. This approach could help improve the NHS's productivity, enhance patient outcomes, and ensure that new technologies are both effective and widely accepted.

4 Issue Tree and Problematization

An issue tree is in general a way to break down the complex task into smaller components for a better approach to address the problem. It helps identify the root causes of a problem and visualize the various factors contributing to it. It helps in systematically analyzing the root causes of a problem and identifying potential solutions. In the context of AI adoption in the NHS, the issue tree categorizes the primary failure to implement AI infrastructure into three key areas: HR challenges, financial constraints, and structural issues. (Hunter, 2023) Each of these categories is further broken down into specific barriers, such as resistance to technological change, poor budget allocation, and inefficiencies in service delivery.



(Designed by Author)

4.1 HR Challenges

One of the major challenges in the adoption of AI within the NHS is resistance to technological change among healthcare professionals. Many NHS staff fear that AI could replace their roles, leading to job displacement. This fear creates reluctance in embracing AI-driven tools, even if they could improve efficiency. Additionally, the successful implementation of AI requires proper training, as healthcare workers must understand how to integrate AI into their workflows. Without adequate training programs, the workforce may struggle to adapt, resulting in the underutilization of AI technology.

4.2 Financial Constraints

Financial constraints also play a crucial role in limiting AI adoption within the NHS. Poor budget allocation means that investment in AI-driven healthcare solutions is not prioritized. The NHS operates within a tight financial framework, and funds are often directed toward immediate operational needs rather than long-term technological advancements. Moreover, inflation further exacerbates the issue by increasing the cost of AI integration, from procuring advanced medical AI software to maintaining and upgrading existing digital infrastructure. Without proper financial planning, the NHS may find it difficult to sustain AI-driven improvements in healthcare services. (Chu, B., 2025.)

4.3 Structural Issues

Structural issues within the NHS also hinder AI implementation. Weak internal controls and governance inefficiencies make it challenging to regulate AI adoption effectively. If there is no clear framework to oversee AI integration, its implementation can become inconsistent and ineffective. Additionally, inefficiencies in service delivery mean that AI solutions, which are designed to streamline patient care, may not be optimally deployed. These structural flaws slow down the potential benefits of AI, preventing it from significantly improving healthcare outcomes. Addressing governance weaknesses and enhancing internal controls is essential for AI-driven transformation in the NHS.

The main problems identified in the issue tree are that although the UK government aims to boost NHS productivity through digitalization, the failure to adopt AI infrastructure raises concerns about inefficiency and financial losses.

5 Findings and Discussions

A structured and evidence-based approach is so important to ensure a successful integration of AI within the NHS. The McKinsey 7-step problem-solving framework provides a logical methodology for analysing complex challenges and formulating effective solutions. By applying this framework, NHS decision-makers can systematically assess AI's impact, identify key issues, and develop a well-structured implementation strategy (Chmielewska et al., 2022).

5.1 Why AI Adoption is Lagging

The NHS faces significant challenges in the adoption of AI despite of their potential. The primary obstacles include poor digital infrastructure that lacks interoperability, making seamless AI integration difficult. Additionally, there is considerable resistance among healthcare professionals who fear job displacement and lack adequate training in AI applications. Financial constraints further exacerbate the issue, with limited budgets and inflationary pressures restricting investment in innovative technologies. Compounding these problems are regulatory hurdles that slow down the approval and implementation of AI

solutions. When compared to other developed healthcare systems like those in the U.S. and Germany, where AI adoption rates are significantly higher, the NHS lags due to these systemic barriers. Overcoming these challenges requires a strategic approach that tackles technological, cultural, and financial bottlenecks simultaneously.

5.2 Strategic Areas for Successful AI Implementation

To resolve this, the NHS must prioritize modernizing its digital backbone through cloud-based platforms and standardized data protocols. Simultaneously, establishing a dedicated AI governance body would ensure ethical compliance, minimize algorithmic biases, and maintain public trust. (Health Research Authority, 2023) By first implementing AI in high-impact areas like triage and radiology, the NHS can demonstrate quick wins while building the foundation for broader adoption. This dual focus on technological upgrades and robust oversight is critical for transforming structural weaknesses into strengths. The Issue Tree underscores that AI adoption in the NHS is not merely a technological shift but a cultural, financial, and structural transformation. By addressing HR resistance through education, securing funding via innovative models, and fixing governance gaps, the NHS can turn AI from a perceived risk into a catalyst for sustainable healthcare innovation.

6 Actionable Recommendations and Conclusions

Based on the analysis done previously using McKinsey 7-step framework, below are the actionable recommendations to address the challenges defined in my issue tree. By focusing on these key areas, the NHS can accelerate AI adoption while ensuring financial sustainability, ethical compliance, and workforce readiness. Through strategic investment, structured training, and transparent patient engagement, the NHS can modernize its operations, improve healthcare outcomes, and enhance overall productivity in the long term.

6.1 Strengthening Technology Infrastructure

To effectively integrate AI within the NHS, upgrading its outdated IT infrastructure is essential. Many NHS Trusts operate on legacy systems that do not support seamless AI integration, leading to inefficiencies and fragmented adoption. A transition to cloud-based and interoperable IT infrastructure will enable AI solutions to work cohesively across different departments and facilities. Additionally, developing a centralized AI platform will standardize data sharing, ensuring that AI-driven insights and patient records are consistent and accessible across all NHS Trusts. AI implementation should begin with high-impact applications, such as radiology, administrative automation, and predictive patient management, where proven success can pave the way for broader adoption. Topol, E. (2019).

6.2 Workforce Readiness and Training

The successful implementation of AI in the NHS is dependent on the readiness of its workforce. Currently, many healthcare professionals lack AI literacy, which can slow down adoption and create resistance to new technologies. To address this, structured AI training programs should be introduced across the NHS, equipping medical staff with a fundamental understanding of machine learning, AI ethics, and system usability.

6.3 Addressing Regulatory and Ethical Barriers

The regulatory landscape surrounding AI in healthcare presents challenges that can slow down its implementation. The NHS must work closely with regulatory bodies to streamline approval processes for AI-driven medical applications, reducing bureaucratic delays while maintaining rigorous safety and ethical standards. Establishing a national AI ethics board can provide governance over AI adoption, ensuring compliance with GDPR, patient data protection laws, and AI accountability measures. Regular AI audits should also be conducted to assess performance, mitigate risks, and address any unintended consequences associated with AI deployment in NHS workflows.

6.4 Ensuring Financial Sustainability and AI Investment

One of the biggest hurdles to widespread AI adoption in the NHS is funding. AI implementation requires significant initial investment in infrastructure, software, and training, which can be difficult given the NHS's already stretched financial resources. To overcome this challenge, the NHS must secure a combination of government funding and private-sector investment through strategic public-private partnerships. In 2022 alone, $19.9bn was spent on hospital goods in the UK. But if such an amount is invested on developing digital infrastructure empowered by AI… it would make a huge impact. Additionally, a structured cost-benefit analysis framework should be implemented to evaluate the return on investment for AI projects, ensuring that NHS resources are allocated effectively. A phased implementation approach should be taken, starting with low-cost, high-impact AI applications such as administrative automation and AI-assisted triage systems before scaling up to more advanced AI-driven diagnostics and predictive analytics.

6.5 Enhancing Public Trust and Patient Engagement

Public trust is a significant factor in the adoption of AI-driven healthcare solutions. Many patients remain sceptical about AI's role in medical decision-making, fearing depersonalization of care and potential data privacy risks. To counter these concerns, the NHS must launch transparent public awareness campaigns that educate patients on the benefits of AI while clearly outlining the measures in place to protect their data. Additionally, AI-powered tools, such as virtual assistants for appointment scheduling, symptom checking, and patient inquiries, can improve accessibility and engagement. Establishing an ethical AI framework will also help maintain trust by ensuring transparency in AI-driven decisions, allowing patients to understand when and how AI is used in their healthcare journey.

7 References

- Government Digital Service (2023) *The founding of the NHS: 75 years on.* Available at: https://history.blog.gov.uk/2023/07/13/the-founding-of-the-nhs-75-years-on/
- University of Huddersfield (2018) *What was healthcare like before the NHS?* Available at: https://blogs.hud.ac.uk/academics/2018/august/whatwashealthcarelikebeforethenhs/
- Primary Care Respiratory Society (2022) *A guide to current NHS architecture across the UK nations*. Available at: https://www.pcrs-uk.org/sites/default/files/2022-December-PCRU-NHS_Architecture.pdf
- Health Foundation (2023) *AI in health care: what do the public and NHS staff think?* Available at: https://www.health.org.uk/reports-and-analysis/analysis/ai-in-health-care-what-do-the-public-and-nhs-staff-think
- Mural (2023) *How to understand your clients' needs and challenges*. Available at: https://www.mural.co/blog/client-needs
- Hunter, D.J. (2023) *At breaking point or already broken? The National Health Service in the United Kingdom*. *New England Journal of Medicine*, 389(2), pp.100–103. Available at: https://www.nejm.org/doi/full/10.1056/NEJMp2301257
- BBC News (2023) *NHS at 75: How has the health service changed?* Available at: https://www.bbc.co.uk/news/articles/cwy7zvp5xrqo
- Chmielewska, M., Stokwiszewski, J., Markowska, J. and Hermanowski, T. (2022) *Evaluating organizational performance of public hospitals using the McKinsey 7-S Framework*. *BMC Health Services Research*, 22(1), Article 7. Available at: https://link.springer.com/article/10.1186/s12913-021-07402-3
- Health Research Authority (2020) *What's next for health research?* Available at: https://www.hra.nhs.uk/about-us/news-updates/whats-next-health-research/
- Health Education England (2019) *The Topol Review: Preparing the healthcare workforce to deliver the digital future*. Available at: https://topol.hee.nhs.uk/

Use of AI-generated content

I acknowledge that:

- ☐ I used ChatGPT/https://chatgpt.com/28 March 2025. to generate materials used just for background research and self-study in the drafting of this submission.

Presentation, structure, and proofreading

I acknowledge that:

- ☐ I did not use AI technology to assist in structuring or presenting my submission.

My final submission has been proofread by

- ☐ Me

_____

Prompts:

- I used ChatGPT to search for some content related to McKinsey 7-step framework.
- I used ChatGPT just to access articles related to the history and background of the NHS.